

MRI Image Processing and Segmentation on Brain Tumor: A Survey

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Abstract— Image segmentation is a critical step in many clinical applications and is crucial in computer-assisted diagnosis. Segmentation of brain is the technique of automatically recognizing malignant brain tissues and labeling them depending on tumor kind. Manual tumor segmentation using brain MRI was time consuming and prone to inaccuracy. In brain MRI research, image segmentation is commonly used to detect and interpret anatomical features, analyze changes to the brain, identify sick areas, organize therapy and image guided surgery. Over the last few decades, different segmentation methods were developed and published in the research, with varying degrees of detail and difficulty. This study will focus on the most common segmentation methods. To begin addressing the complex details and challenges of the MRI Brain issue, we have introduced fundamental image preprocessing stages and then we have performed Image segmentation.

Keywords— Brain tumor, CNN, MRI, OpenCV, Segmentation

I. INTRODUCTION

Image segmentation is essential for various applications such as image comprehension, feature extraction, analysis and interpretation. Some of its medical applications include tissue categorization, tumor identification, tumor volume computing, blood cell segmentation, therapies, axis fitting, and image verification are all part of the process. Statistical methods such as extracting features, analysis, and quantification could be used in images to identify disease, track the progression of sickness, or compare a healthy person to an ill one. Precision and consistency in tumor measurement and morphology are critical for brain tumor diagnosis, treatment, and monitoring response to oncological therapy. Segmentation is the process of distinguishing among tumor and functional brain tissues a brain tumor[1][8]. A brain tumor can be a normal (malignant) and cancerous abnormal cell formation in the brain. A benign tumor has a stable appearance and a small number of cells, even though a malignant brain tumor does have a non-uniform form and a large number of active cells[7]. Low-grade benign tumors include gliomas and meningiomas. They resemble typical cells in the brain and grow slowly. Greater malignant tumors

include astrocytomas and glioblastoma. They have a predisposition towards rapid growth[3]. Brain segmentation is an important step in neurology that has a variety of uses, including data analysis, planning and brain imaging[15]. Despite the fact that MRI can accurately represent brain patterns, segmentation is difficult due to poor pixel size, poor contrast, non-uniformity, partial volume impact, noise, object shape variability, and other procurement artifacts in the extracted data, as well as a lack of human biology models that truly capture the prospective deformation in each framework. The treatment of a brain tumor is determined by patient's condition, tumor category, and position[4]. Tumors can form and spread to nearby healthy tissue, making therapy and diagnosis difficult. As a result, good brain tumor segmentation from deeper structures is crucial for finding tumors early and enhancing individuals' chances of survival. Due to the obvious convoluted process and aesthetic impact of the tumor cells, the tumor boundaries are often hazy, and the tumors may have spread into the nearby area of the brain, hard to distinguish the impacted tissue from another embracing healthy tissue, precise segmentation of neurological disorders from healthy tissue Magnetic resonance imaging is a complicated and challenging process. As a result, manually recognizing tumor borders (delineation) in MRI images takes time and is prone to inaccuracy[11]. Automated brain tumor segmentation with MRI images tackles these challenges by precisely determining the type and exact location of tumors[17], facilitating a quick and accurate diagnosis. Tumor that are treated quickly may help the patient recover.

II. DATA PREPROCESSING

OpenCV includes a number of filtering algorithms for performing picture smoothing tasks. Smoothing methods are commonly used to minimize noise, detail, and so forth. These approaches can also be used to decrease pixelation in low-resolution images[5][14]. Image filtering is used to remove noise, increase the contrast and edge detection.

In OpenCV, these are the frequently used image filters:

- Gaussian filter
- Averaging filter

- Bilateral filter
- Median filter

Gaussian filter

cv2 function is included in the OpenCV library Gaussian filtering. The GaussianBlur() function blurs a picture using a Gaussian Kernel. The Gaussian Kernel multiplies each pixel in a picture. A Gaussian Kernel is essentially a based clustering of pixels. When we employ the Gaussian Filter on to an image, we first set the value of the Kernel/Matrix which will be used to degrade the picture. Sizes are usually odd, which means that the total results may be calculated on the centre pixel.

TABLE I PARAMETERS OF GAUSSIAN FILTER

Name	Description
src	Input picture
ksize	Gaussian kernel size (width and height)
sigmaX	The standard deviation of the Gaussian kernel along the X-axis
dst	Output picture
sigmaY	The standard deviation of the Gaussian kernel along the Y-axis

Averaging filter

The picture is the source of the blurring, and ksize is the length of the blurring kernel. The image's origin is indicated by the src (source) attribute, destination image dst has the same dimensions as the source image. The anchor coordinates are shown by the anchor. It is arranged to basic point (coordinate (-1,1)) by default. ksize argument gives the length of the blurring kernel. The category of border used is stored in the borderType, which is a number object.

Bilateral filter

To blur an image with a bilateral filter, use the cv2.bilateralFilter() method in Python OpenCV. This function may be used to minimize noise while maintaining sharp edges.

TABLE II PARAMETERS OF BILATERAL FILTER

Name	Description
image	Source of the image
dst	Destination Image
d	Pixel neighborhood's diameter
sigmaColor	Sigma's value in the color space
sigmaSpace	Sigma's value is in the coordinate space

Median filter

The cv2.medianBlur() function in Python OpenCV is used to smudge an image with a median kernel. This is an algorithm for nonlinear filtering. It's really efficient in getting rid of salt-and-pepper noises. This replaces the center component with the average of all the pixel under the kernel region. Because we choose the middle, the resulting

image will include no other pixel values than those from the actual image. The image is the operation's picture, and the ksize is a length object reflecting the kernel's size.

III. SEGMENTATION METHODS

Thresholding

The threshold approach is perhaps the most fundamental visual segmentation method. By measuring the ratio of every pixel to a specified value, it distinguishes pixels in a visual (threshold). When the intended object is more apparent than the background (unnecessary components), it is beneficial [9][10]. It is possible to consider the threshold (T) constant, and it would only operate if there was no noise in the image.. One may keep this threshold value fixed or changeable, depending on your needs. The thresholding method divides a greyscale image into needed and not-needed areas in order to convert it to a binary image.

Edge-Based

One of the most common segmentation approaches is edge-based. It identifies the borders between distinct objects in a picture. This is a very important step, since borders contain a lot of useful information that we can use to figure out the properties of different things in an image[2]. Edge detection being commonly used which makes it easier to remove extraneous and undesired data from imagery. It substantially decreases the image's size, allowing it much simpler for analyze. Texturing, brightness, grey tone, color, intensity, and other image properties are used by edge-based algorithms to find an image's edges.

Region Based

Using region-based algorithms, the image is divided among parts of equal quality. The method finds those categories first by selecting a preprocessing step, which could be a small or big slice of an input images. A region-based strategy will either add extra pixels to the initial centroids or decrease them such that when identified, they would mix with another centroid.

Watershed Segmentation

In Image processing, this segmentation is indeed a monochromatic image alteration. To interpret the image as though it were a spatial terms, a watershed method would be utilized. It looks for lines run all along top of the ridge and uses the intensity of a pixel to determine their length [6][12]. The term "watershed" has a number of technical meanings and applications. In addition to recognizing the ridges of such pixels, it focuses on creating basins and filling them with markers till they approach those watershed lines of ridges. Using the watershed method, each image is transformed into a topographic map. The grey levels of their pixels would represent the geography of the watershed segmentation technique.

Clustering-Based Segmentation

They are unsupervised techniques that let you find hidden data in an image that you cannot see with your eyes. Among

the hidden data are clusters, patterns, shadings, and other information. As the name indicates, a clustering approach divides a picture into groups of pixels with common features. This would split the input in regions, having items in one region being far more similar to those in another.

Neural Networks for Segmentation

AI is being used to analyze a photograph and recognize its many elements, including such individuals, word and objects.

CNN are widely used for segmenting the image because of its ability to recognize and interpret image information rapidly and easily [8]. Mask R-CNN is indeed a framework that can generate a pixel-by-pixel mask by each item inside a image. It is a more advanced version of the Accelerated R-CNN object detection framework.

Generative adversarial networks (GANs) are a type of system that contains unsupervised learning to train two models at the same time. Generative adversarial networks are designed to simultaneously train two deep learning models. There are two models in these networks: (G) generator and (D) discriminator.

RNN, which stand for "recurrent neural network," are a subcategory of "deep learning" networks [19]. Because of their capacity to process serial data, these networks have performed well in situations where the data sequence is critical. RNNs have just a monotonous task for each element of a sequence; because the output of this network is tied to the computations that came before it, the results are referred to as recurrent.

The upgraded form of RNNs created for sequence data is known as long short-term memory, or LSTM for short. Each LSTM unit has a view of a pixel and takes in information from LSTM units that came before it [20]. In this approach, it collects data in an iterative manner using all of the pixel intensities that are present in the image.

A deep belief network, also known as a DBN, is in point of fact a connection that has been built between a wide variety of brain systems that are linked together in a variety of series configurations. Deep belief networks that generate output are a type of learning that can only be achieved through unsupervised practice [13][18]. They are composed of binary latent constructs and have layers that can either be undirected or directed in nature. Each layer of deep belief networks is responsible for learning the entirety of the input

IV. LITERATURE REVIEW

In [1] this research, the author employed CNN to separate the tumor on MRI images, by the first stage using contrast leveling from many detectors. Tumor segment is accomplished in this study depending on tumor stage, whether this is High or low grade. Tiny kernels are being utilized to investigate the feasibility of deep networks. This is accomplished by contrasting deep CNN architectures with shallow ones.

The study [2] examined images acquired at various ages and using various acquisition processes. In addition, several patch sizes are used, with each patch focused on a different aspect of the segmentation. The smaller area enables more in-depth research, but it lacks organization is to maximize. The bigger patch, on the other side, generates smoother segmentation but fails to capture finer features.

A [3] MU-Net is presented for Brain Tissue Identification and Segmentation. The rasterization is deconvolved rather than convolved in U-Net. In this work, the deconvolution layer is substituted with an up-sampling layer, which is followed by two convolution operations, an additional up-sampling layer, and elastic transformation enhancement.

This [4] paper presents a basic introduction that will assist the reader in becoming a specialist in automatic segmentation. The first step is to separate brain tumors from normal brain tissue in using statistical image feature assessment for prognosis as well as therapy planning, which might include the use of mammography. This is why the severe significance of Glioblastoma multiforme is so important.

In this [5] research k-means clustering algorithm is proposed. The computation of area during brain tumor segmentation is easy, and the specific location of the tumor is identified once it has been segmented. The effectiveness of fuzzy c-means is critical as compared to other methods.

This [6] paper investigated several automated tumor segmentation methods for MRI, CT, PET, and multimodal imaging. To give insight into various solutions, the technique, benefits, limits, and potential difficulties are discussed. Tumor segmentation approaches are becoming increasingly popular owing to the significant tissue differences and benign nature of MRI.

V. IMAGE PROCESSING STAGES

The algorithm is divided into two stages: pre-processing of a given MRI picture, segmentation, and subsequently morphological operations.

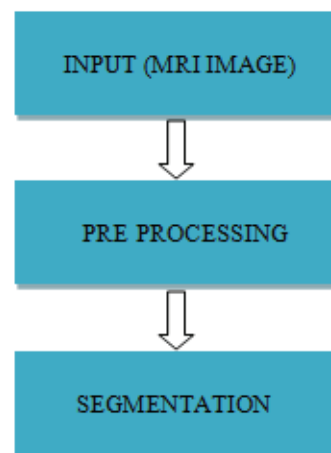


Fig.1:Image processing steps

The algorithm steps are as follows:

1. Provide an MRI image of the brain as data.

Here a MRI image [16] has been taken from the BraTS 2018 and used as a input image. To load and read the picture as a Numpy Array, we may use the method `cv2.imread()`.

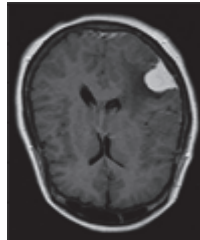


Fig. 2: Input image

2. Change the image to grayscale.

The Input image needs to be converted to Gray scale image using `cv2.cvtColor()`.

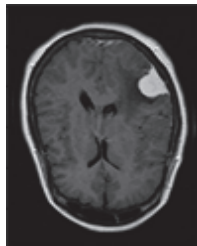


Fig. 3: Grayscale Image

3. Perform threshold segmentation.

The image was transformed into grayscale, and the resultant image was utilized for thresholding. Thresholding is a technique in OpenCV that assigns pixel values according to a specified threshold level. Every pixel data is converted to the threshold value during thresholding if indeed the pixel value is much less than the cutoff, it is set to zero; otherwise, it is changed to the highest value. Image operation on thresholding can be performed using `cv2.threshold()`

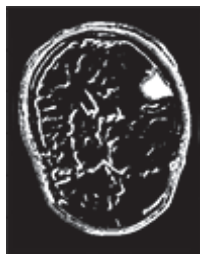


Fig.4: Threshold Image

4. Perform the morphological procedure.

Morphological transformations are straightforward procedures based on picture shape. They are often used to binary pictures. The required inputs consist of the original image

and a structural element, often known as a kernel, which dictates the type of operation to be performed.

The morphological operators applied are erosion and dilation. The borders of foreground items are eroded by erosion. A pixel in the main image (whether 1 or 0) is considered 1 if all pixels beneath the kernel are likewise 1. Otherwise, it will deteriorate. As a result, depending on size of the kernel, pixels near the border will be eliminated. As a result, the depth or size of the front item, or simply the white zone in the image, diminishes. As a result, we utilized this to eliminate the little white noise. To remove the noise using Morphological, we need both `cv2.morphologyEx()` and then `cv2.dilate()`.



Fig.5: Morphological Image

5. Determine the watershed segmentation.

The Watershed Algorithm was used to segment the data. OpenCV designed a water-shedding technique based on markers, which is an active image segmentation approach. It assigns multiple labels to objects by marking the areas we are confident about with one color (or eroded with another color), and the regions we are uncertain about with 0. That is our point of reference.



Fig.6: Final Segmented Image

The watershed algorithm is then used. Next the marker will be modified with marks we provided, and the object bounds would have a score of -1. This is essentially a region-building strategy.

CONCLUSION

Brain Tumor Segmentation in image data using different segmentation methods has been explored. Future advancements in tumor evaluation approaches, including cancer growth calculation, and multiclass tumor categorization, will enhance existing systems. Brain tumor segmentation methodologies, together with all the previously mentioned amazing breakthroughs in this sector, will undoubtedly show substantial promise at later stage. The difficulty of image segmentation in medical imaging

entails separating organisms and malignancies from medical data. For brain tumor segmentation using Watershed segmentation has proven accurate results. MRI images can be further classified using machine learning algorithms like SVM, Random Forest and Logistic Regression.

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REFERENCES

- [1] Pereira, S., Pinto, A., Alves, V., & Silva, C. A. (2016). Brain tumor segmentation using convolutional neural networks in MRI images. *IEEE transactions on medical imaging*, 35(5), 1240-1251.
- [2] Moeskops, P., Viergever, M. A., Mendrik, A. M., De Vries, L. S., Benders, M. J., & Išgum, I. (2016). Automatic segmentation of MRI brain images with a convolutional neural network. *IEEE transactions on medical imaging*, 35(5), 1252-1261.
- [3] Hasan, S. K., & Linte, C. A. (2018, October). A modified U-Net convolutional network featuring a nearest-neighbor re-sampling-based elastic-transformation for brain tissue characterization and segmentation. In 2018 IEEE Western New York Image and Signal Processing Workshop (WNYISPW) (pp. 1-5). IEEE.
- [4] Bhandari, A., Koppen, J., & Agzarian, M. (2020). Convolutional neural networks for brain tumour segmentation. *Insights into Imaging*, 11(1), 1-9.
- [5] Jose, A., Ravi, S., & Sambath, M. (2014). Brain tumor segmentation using k-means clustering and fuzzy c-means algorithms and its area calculation. *International Journal of Innovative Research in Computer and Communication Engineering*, 2(2).
- [6] Angulakshmi, M., & Lakshmi Priya, G. G. (2017). Automated brain tumour segmentation techniques—a review. *International Journal of Imaging Systems and Technology*, 27(1), 66-77.
- [7] Ramasamy, J., Doshi, R., & Hiran, K. K. (2021, August). Segmentation of Brain Tumor using Deep Learning Methods: A Review. In *Proceedings of the International Conference on Data Science, Machine Learning and Artificial Intelligence* (pp. 209-215).
- [8] Hatamizadeh, A., Tang, Y., Nath, V., Yang, D., Myronenko, A., Landman, B., & Xu, D. (2022). Unetr: Transformers for 3d medical image segmentation. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision* (pp. 574-584).
- [9] Fayjie, A. R., Dutta, R., Kashyap, P., Kumar, U. R., & Vandewalle, P. (2022). Semi-Supervised Adversarial Few-Shot Learning for Medical Image Segmentation.
- [10] Das, S., Nayak, G. K., Saba, L., Kalra, M., Suri, J. S., & Saxena, S. (2022). An artificial intelligence framework and its bias for brain tumor segmentation: A narrative review. *Computers in Biology and Medicine*, 105273.
- [11] Huang, Z., Lin, L., Cheng, P., Peng, L., & Tang, X. (2022). Multi-modal Brain Tumor Segmentation via Missing Modality Synthesis and Modality-level Attention Fusion. *arXiv preprint arXiv:2203.04586*.
- [12] Alqazzaz, S., Sun, X., Nokes, L. D., Yang, H., Yang, Y., Xu, R., & Yang, X. (2022). Combined features in region of interest for brain tumor segmentation. *Journal of Digital Imaging*, 1-9.
- [13] Ranjbarzadeh, R., Bagherian Kasgari, A., Jafarzadeh Ghouschi, S., Anari, S., Naseri, M., & Bendechache, M. (2021). Brain tumor segmentation based on deep learning and an attention mechanism using MRI multi-modalities brain images. *Scientific Reports*, 11(1), 1-17.
- [14] Rao, C. S., & Karunakara, K. (2021). A comprehensive review on brain tumor segmentation and classification of MRI images. *Multimedia Tools and Applications*, 80(12), 17611-17643.
- [15] Khan, A. R., Khan, S., Harouni, M., Abbasi, R., Iqbal, S., & Mehmood, Z. (2021). Brain tumor segmentation using K-means clustering and deep learning with synthetic data augmentation for classification. *Microscopy Research and Technique*, 84(7), 1389-1399.
- [16] Wadhwa, A., Bhardwaj, A., & Verma, V. S. (2019). A review on brain tumor segmentation of MRI images. *Magnetic resonance imaging*, 61, 247-259.
- [17] Abdulbaqi, H. S., Jafri, M. Z. M., Omar, A. F., Mutter, K. N., Abood, L. K., & Mustafa, I. S. B. (2015, December). Segmentation and estimation of brain tumor volume in computed tomography scan images using hidden Markov random field Expectation Maximization algorithm. In 2015 IEEE student conference on research and development (SCoReD) (pp. 55-60). IEEE.
- [18] Le, N., Yamazaki, K., Quach, K. G., Truong, D., & Savvides, M. (2021 January). A multi-task contextual atrous residual network for brain tumor detection & segmentation. In 2020 25th International Conference on Pattern Recognition (ICPR) (pp. 5943-5950). IEEE.
- [19] Das, S., Nayak, G. K., Saba, L., Kalra, M., Suri, J. S., & Saxena, S. (2022). An artificial intelligence framework and its bias for brain tumor segmentation: A narrative review. *Computers in Biology and Medicine*, 105273.
- [20] Ramya, P., Thanabal, M. S., & Dharmaraja, C. (2021). Brain tumor segmentation using cluster ensemble and deep super learner for classification of MRI. *Journal of Ambient Intelligence and Humanized Computing*, 12(10), 9939-9952.